Thadomal Shahani Engineering College

Name: Alok Prasad			**Subject: SEPM Laboratory**
TE	**T21**	**91**	**AIDS**

Experiment No: 4

Aim: To understand Continuous Integration, install and configure Jenkins with Maven/Ant/Gradle to setup a build Job

Steps:

To install Jenkins following software packages are required

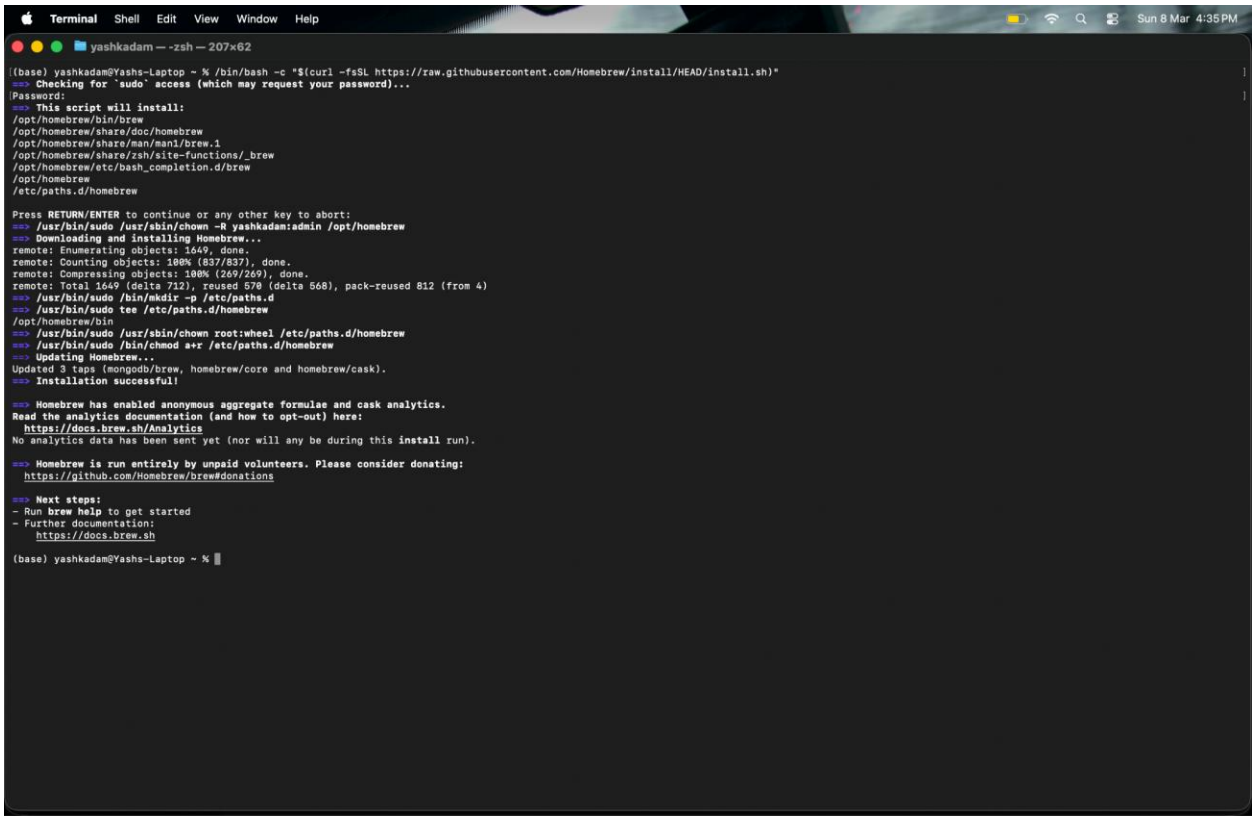
1. GIT (git-scm.com)
2. Notepad++ (https://notepad-plus-plus.org/downloads/)
3. Latest Java development kit (JDK)
4. Jenkins
5. Apache Maven (Optional)

Step 1-: Install GIT
Step 2 -: Install Notepad++
Step 3 -: Install Java
Step 4 -: Install Jenkins
Step 5 -: Install Maven

Jenkins is an open source automation tool written in Java with plugins built for Continuous Integration purpose. Jenkins is used to build and test your software projects continuously making it easier for developers to integrate changes to the project, and making it easier for users to obtain a fresh build. It also allows you to

continuously deliver your software by integrating with a large number of testing and deployment technologies.

Output:



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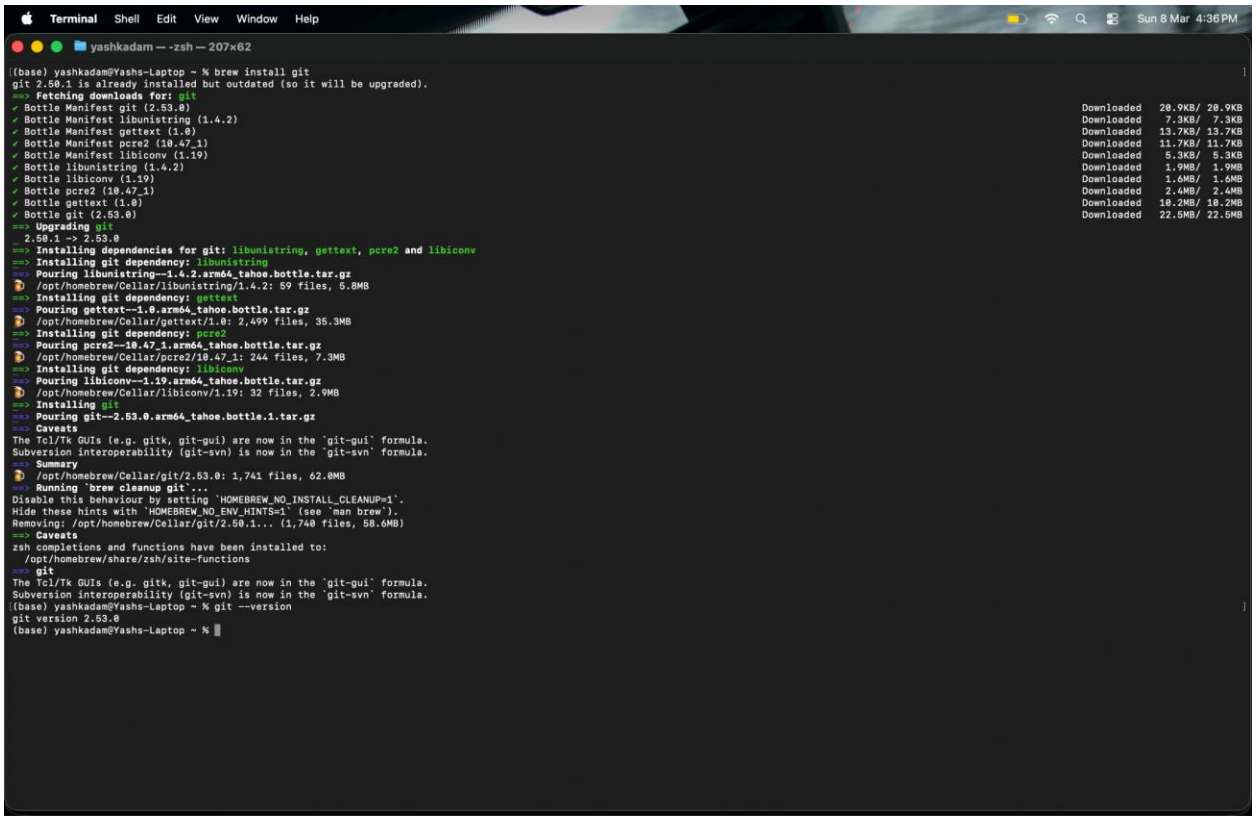

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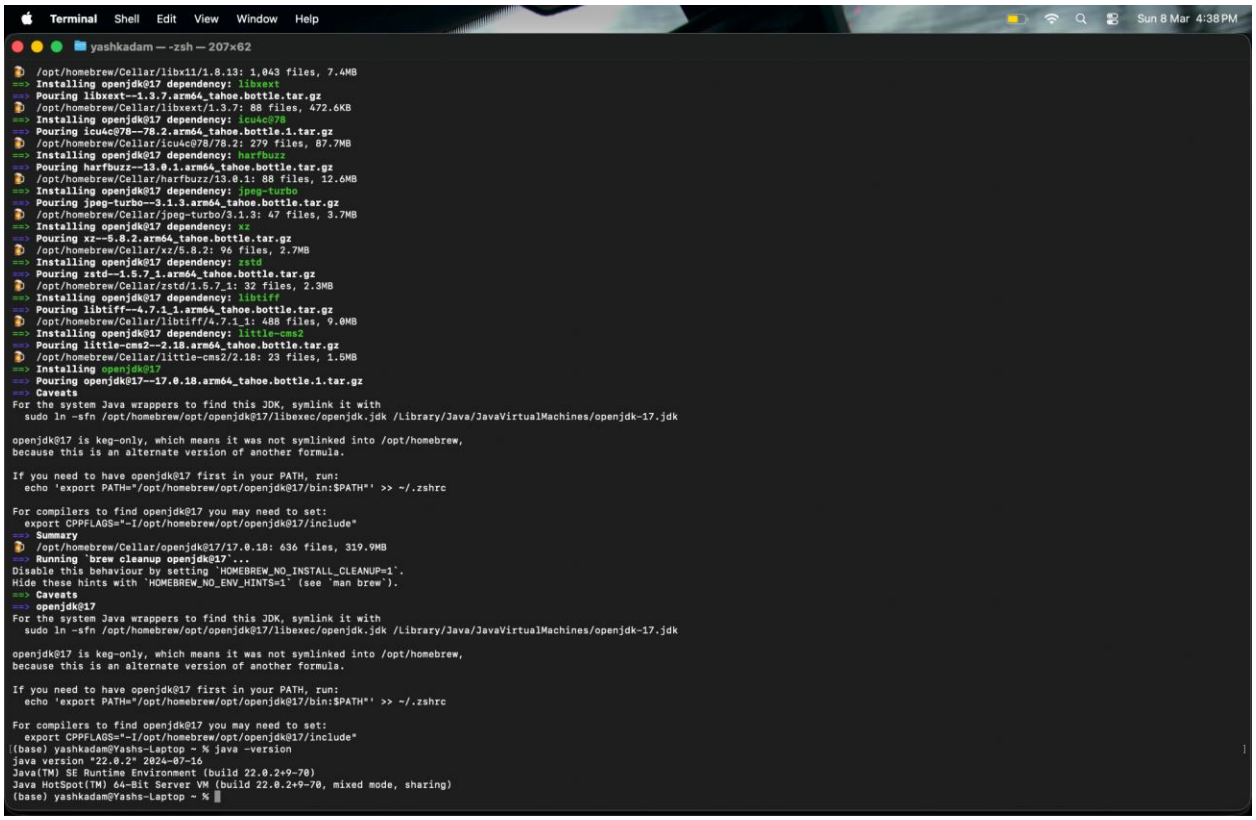

```
Terminal  Shell  Edit  View  Window  Help                                   Sun 8 Mar 4:38 PM
yashkadam — -zsh — 207×62
🍺  /opt/homebrew/Cellar/libx11/1.8.13: 1,043 files, 7.4MB
==> Installing openjdk@17 dependency: libxext
==> Pouring libxext--1.3.7.arm64_tahoe.bottle.tar.gz
🍺  /opt/homebrew/Cellar/libxext/1.3.7: 88 files, 472.6KB
==> Installing openjdk@17 dependency: icu4c@78
==> Pouring icu4c@78--78.2.arm64_tahoe.bottle.1.tar.gz
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==> Installing openjdk@17 dependency: harfbuzz
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==> Pouring jpeg-turbo--3.1.3.arm64_tahoe.bottle.tar.gz
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==> Installing openjdk@17
==> Pouring openjdk@17--17.0.18.arm64_tahoe.bottle.1.tar.gz
==> Caveats
For the system Java wrappers to find this JDK, symlink it with
  sudo ln -sfn /opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk /Library/Java/JavaVirtualMachines/openjdk-17.jdk

openjdk@17 is keg-only, which means it was not symlinked into /opt/homebrew,
because this is an alternate version of another formula.

If you need to have openjdk@17 first in your PATH, run:
  echo 'export PATH="/opt/homebrew/opt/openjdk@17/bin:$PATH"' >> ~/.zshrc

For compilers to find openjdk@17 you may need to set:
  export CPPFLAGS="-I/opt/homebrew/opt/openjdk@17/include"
==> Summary
🍺  /opt/homebrew/Cellar/openjdk@17/17.0.18: 636 files, 319.9MB
==> Running `brew cleanup openjdk@17`...
Disable this behaviour by setting `HOMEBREW_NO_INSTALL_CLEANUP=1`.
Hide these hints with `HOMEBREW_NO_ENV_HINTS=1` (see `man brew`).
==> Caveats
==> openjdk@17
For the system Java wrappers to find this JDK, symlink it with
  sudo ln -sfn /opt/homebrew/opt/openjdk@17/libexec/openjdk.jdk /Library/Java/JavaVirtualMachines/openjdk-17.jdk

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For compilers to find openjdk@17 you may need to set:
  export CPPFLAGS="-I/opt/homebrew/opt/openjdk@17/include"
(base) yashkadam@Yashs-Laptop ~ % java -version
java version "22.0.2" 2024-07-16
Java(TM) SE Runtime Environment (build 22.0.2+9-70)
Java HotSpot(TM) 64-Bit Server VM (build 22.0.2+9-70, mixed mode, sharing)
(base) yashkadam@Yashs-Laptop ~ %
```

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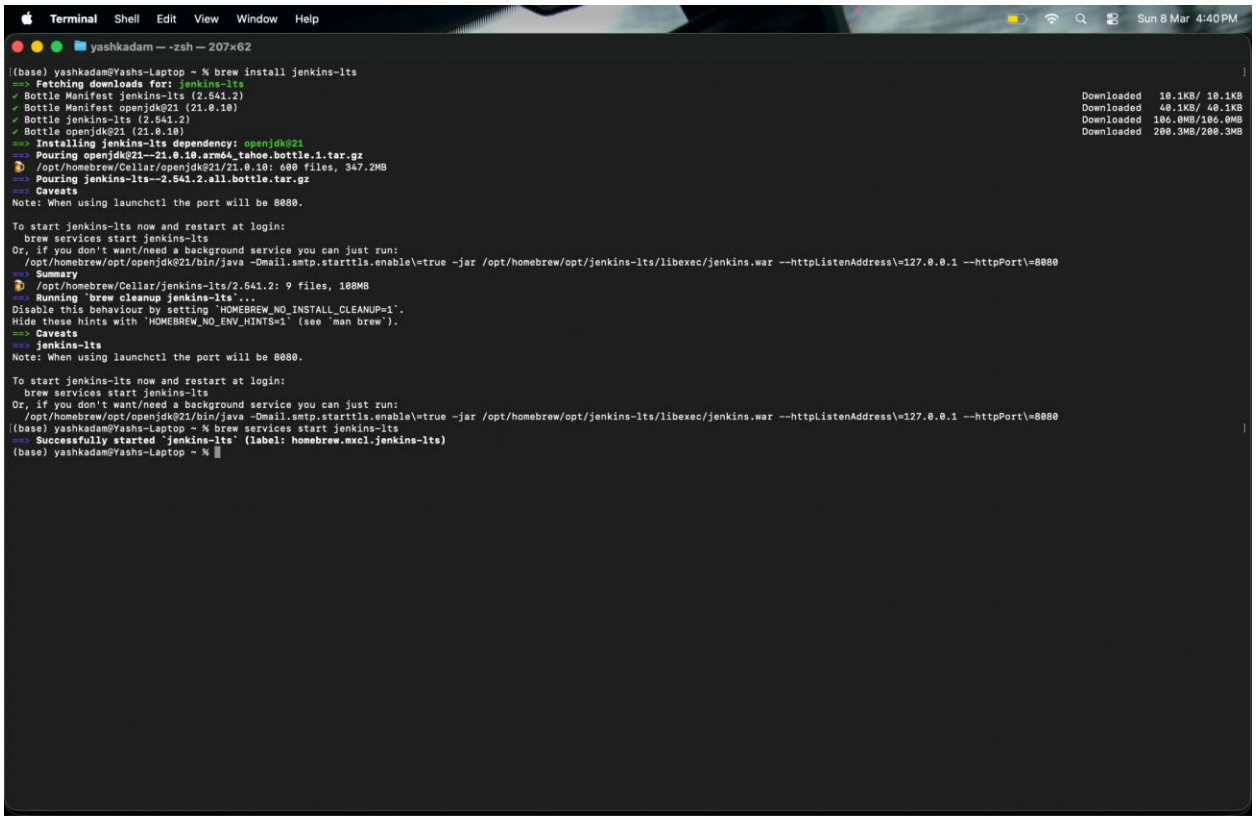

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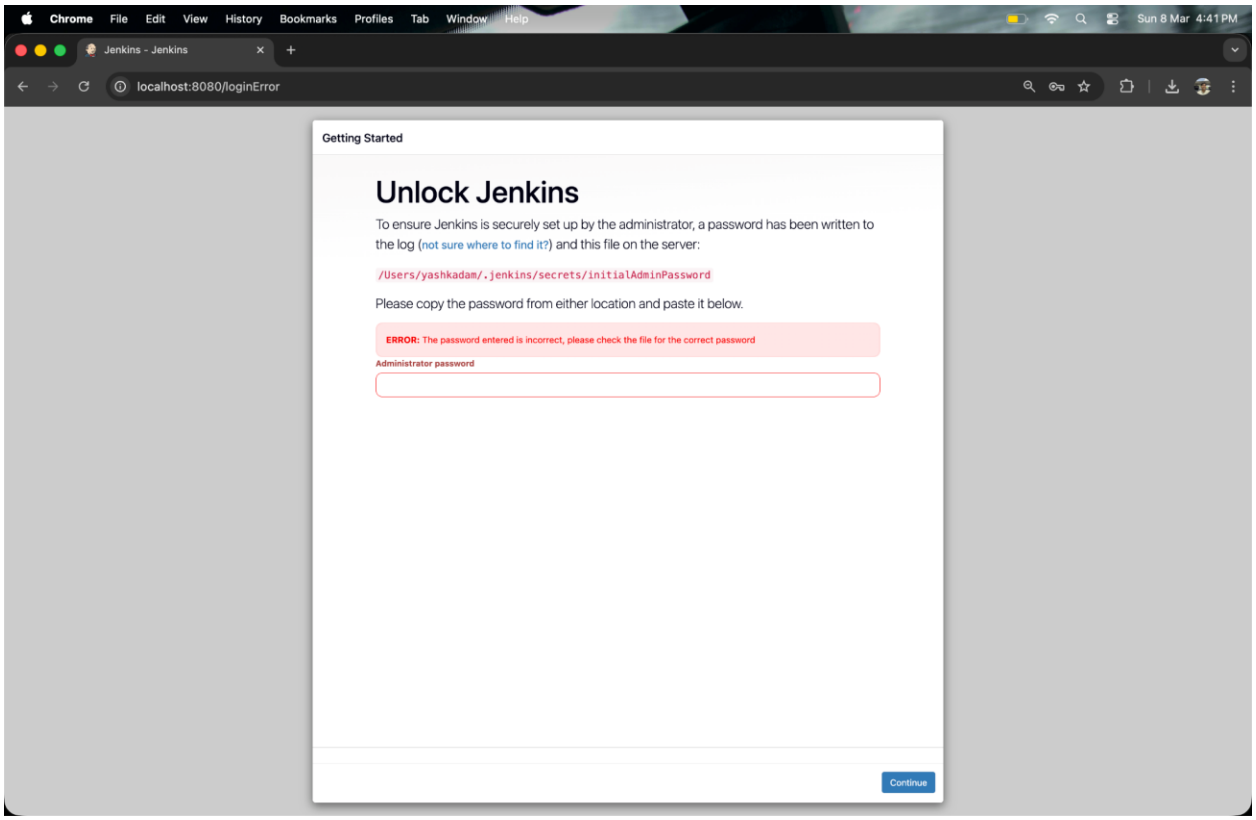

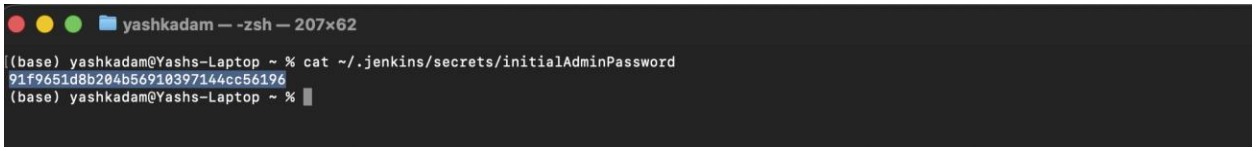

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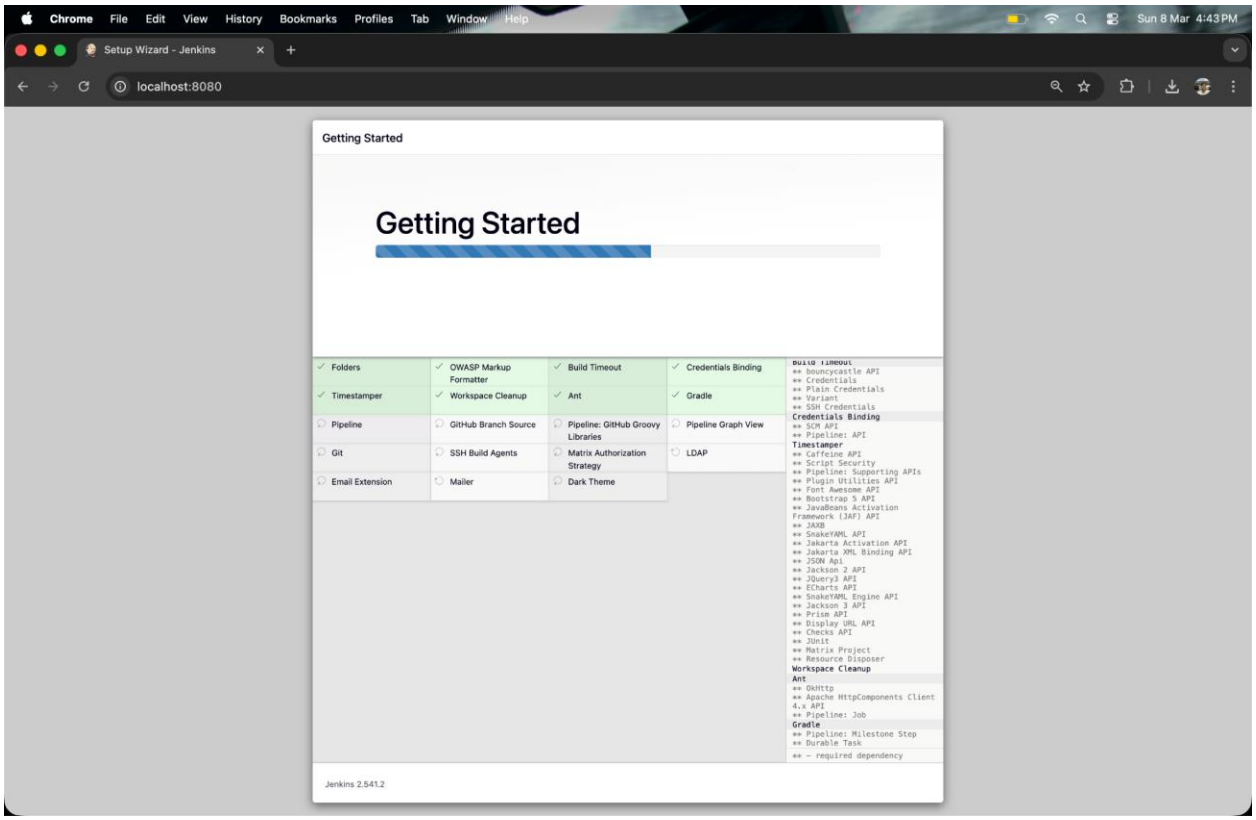

TE -2025-26 Batch: 13 Sem: 6 Roll No:2301052 Date: 09/02/2025

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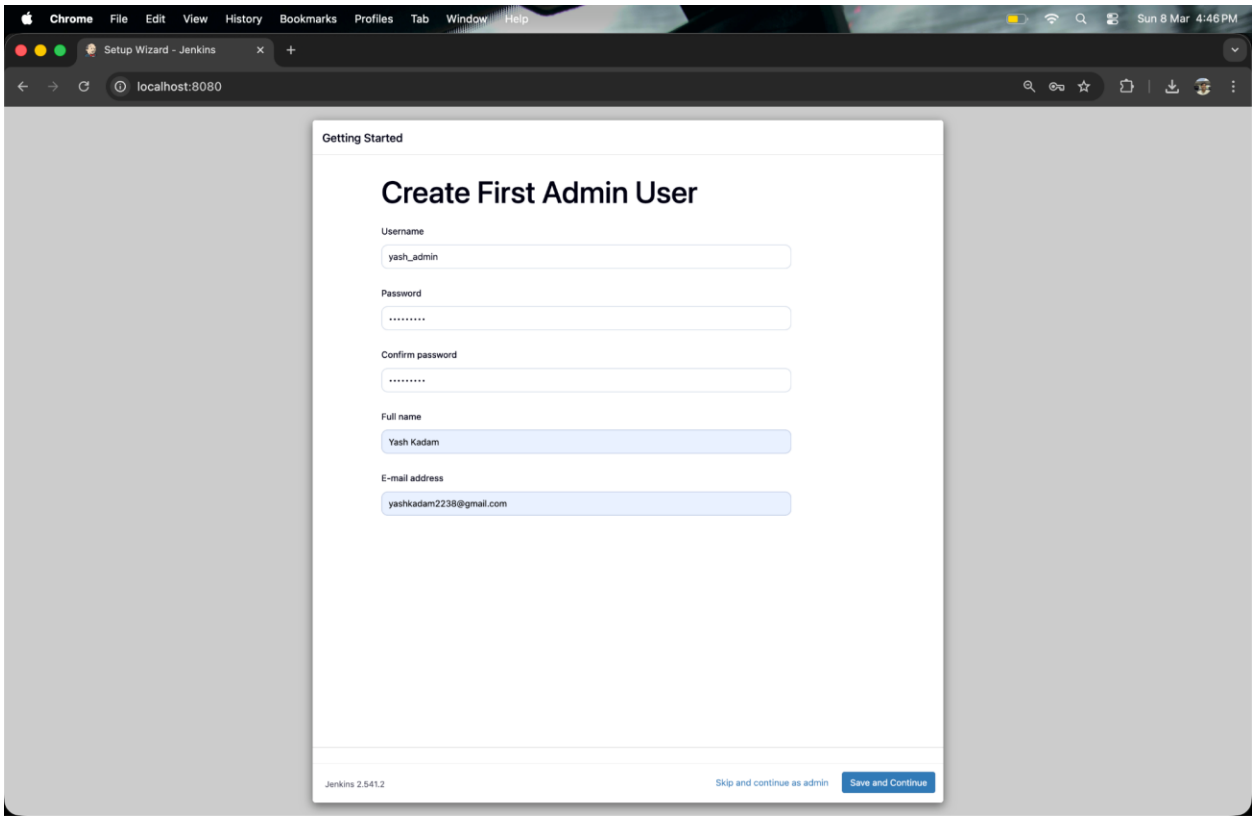

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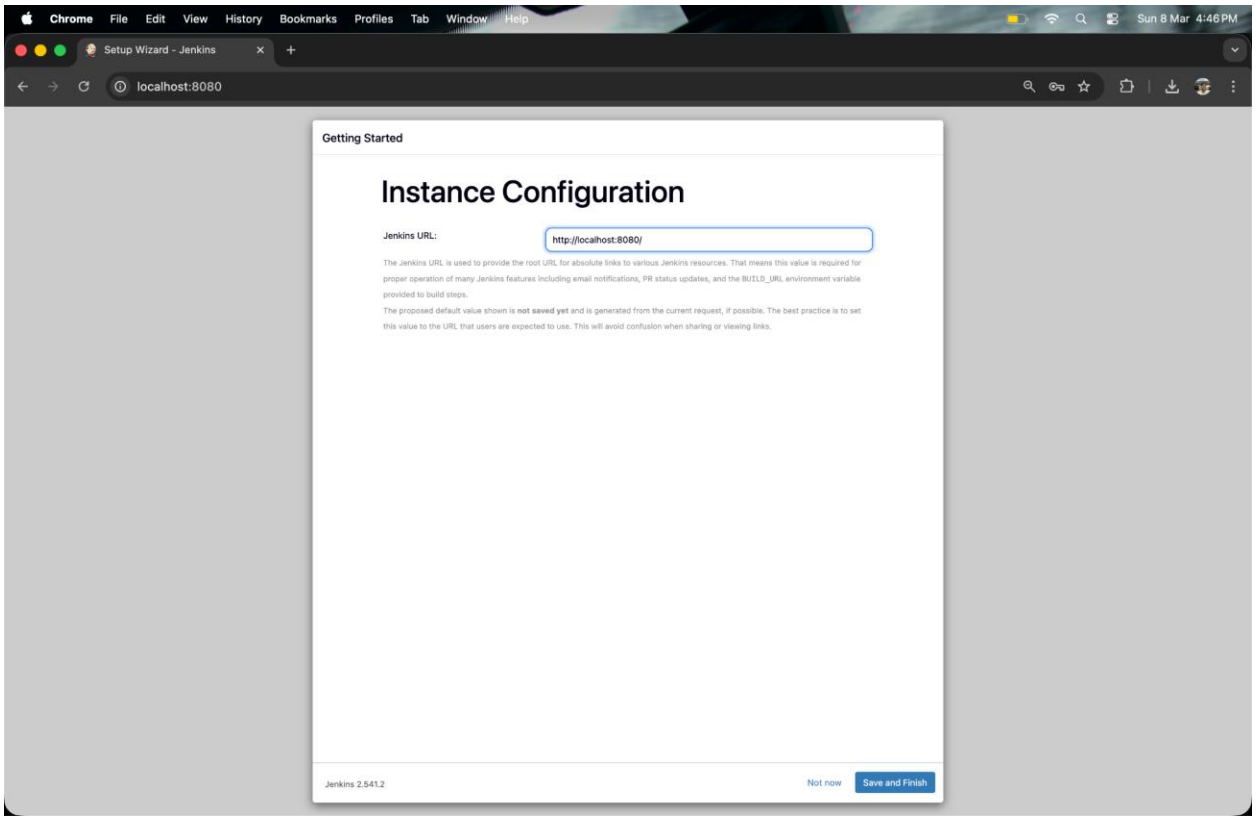

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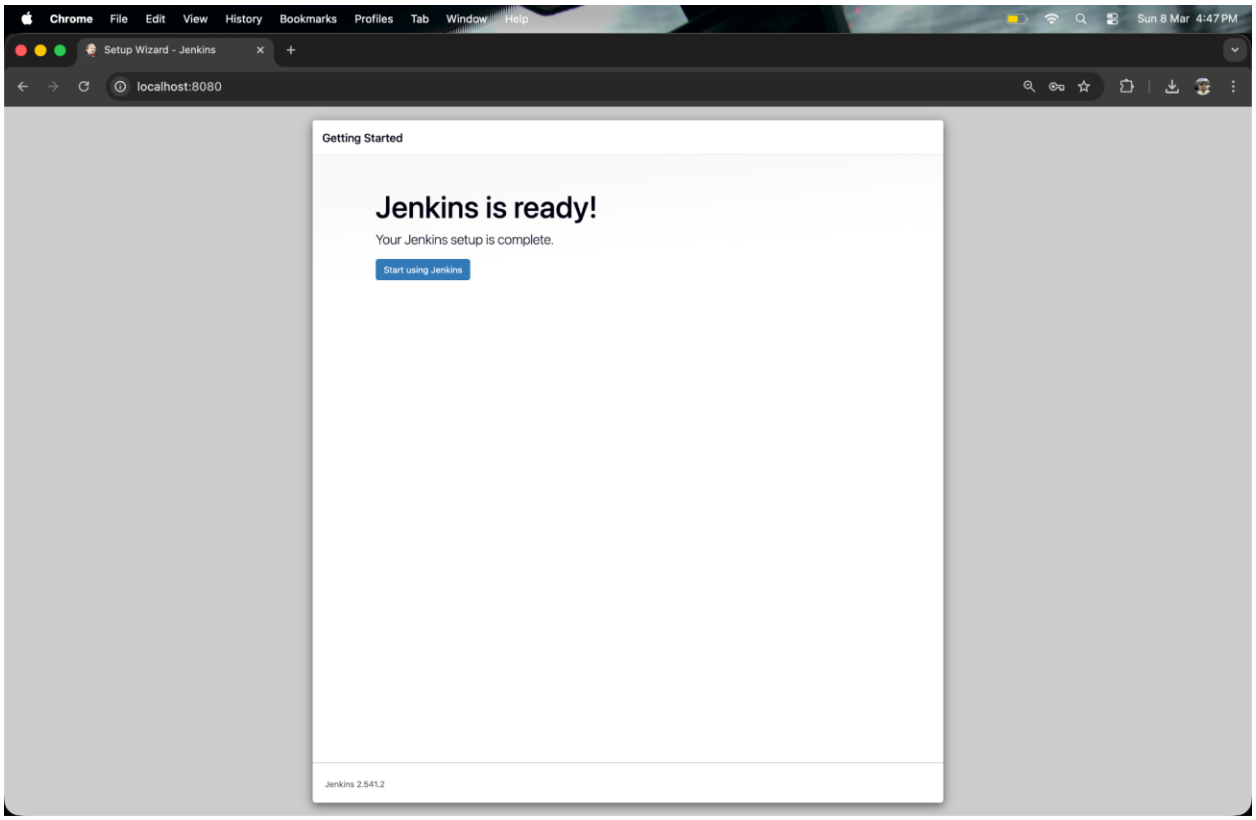

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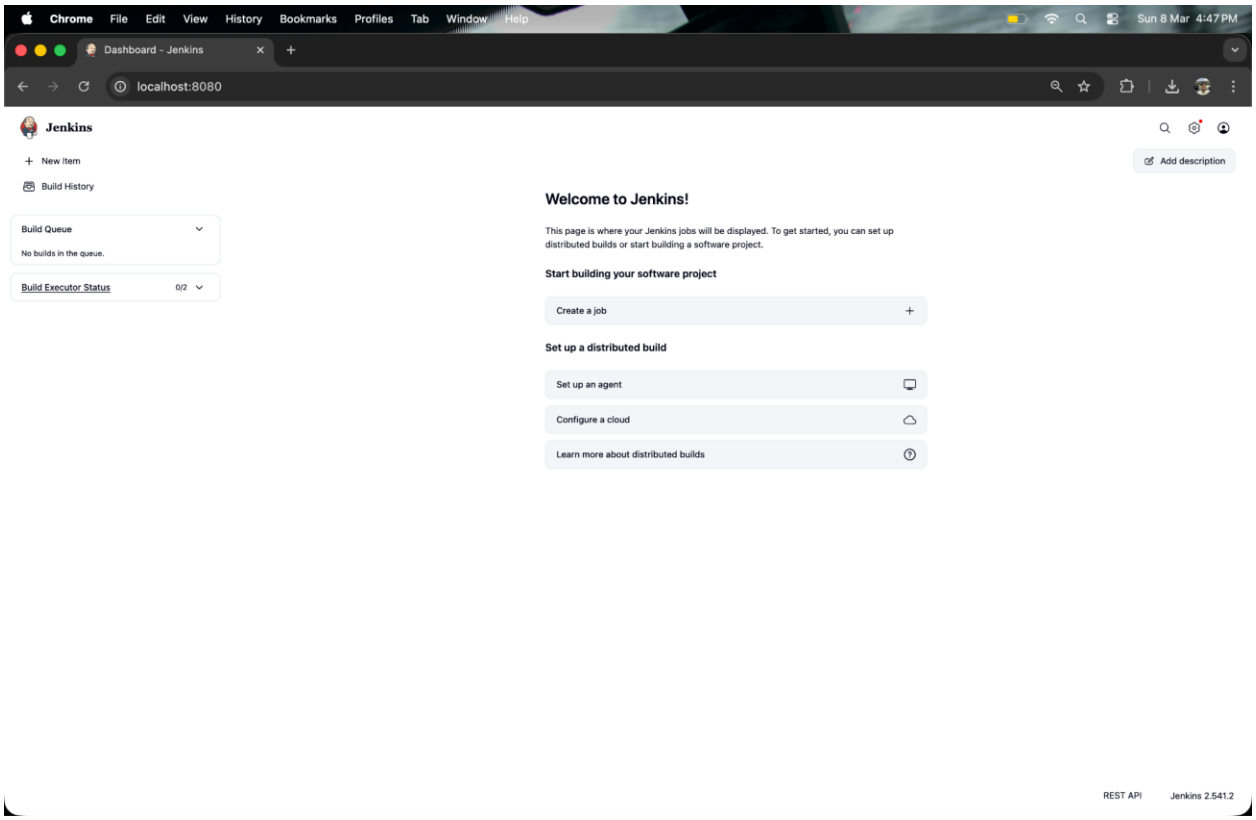

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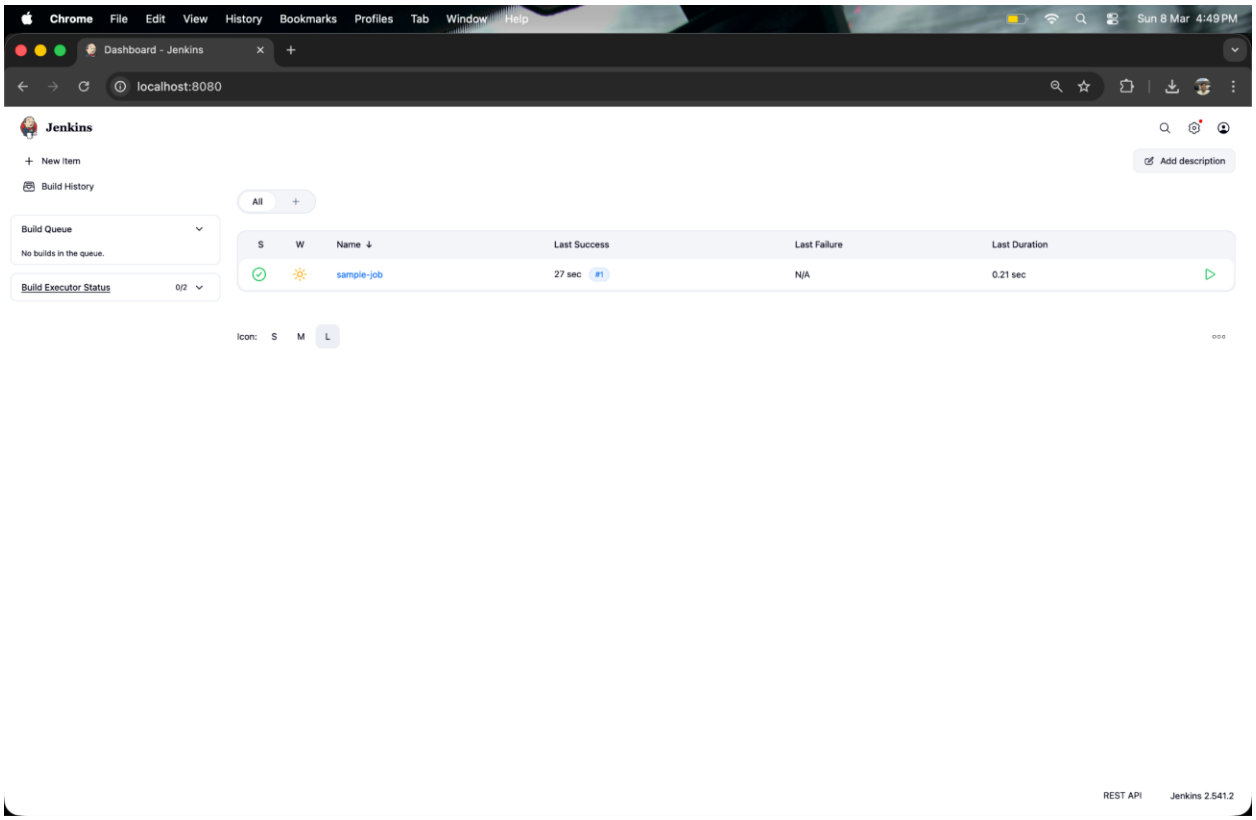

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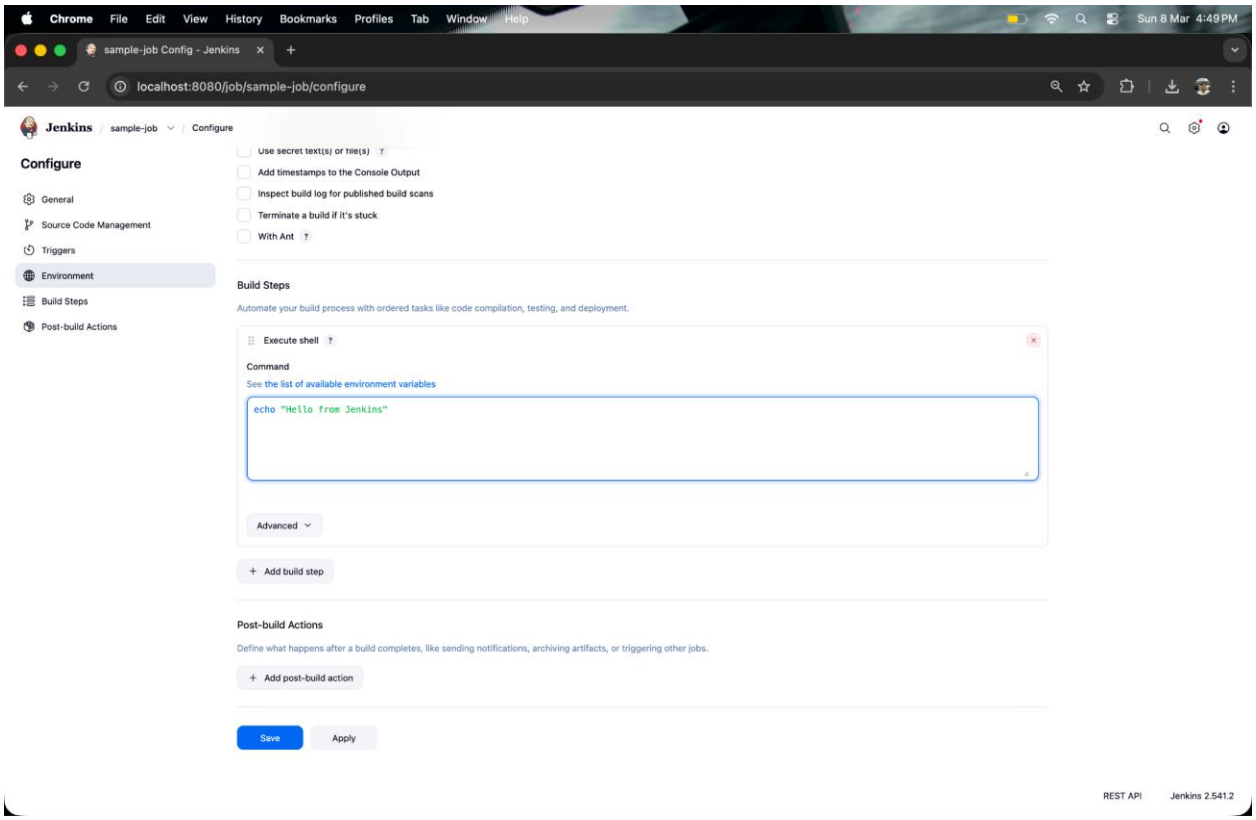

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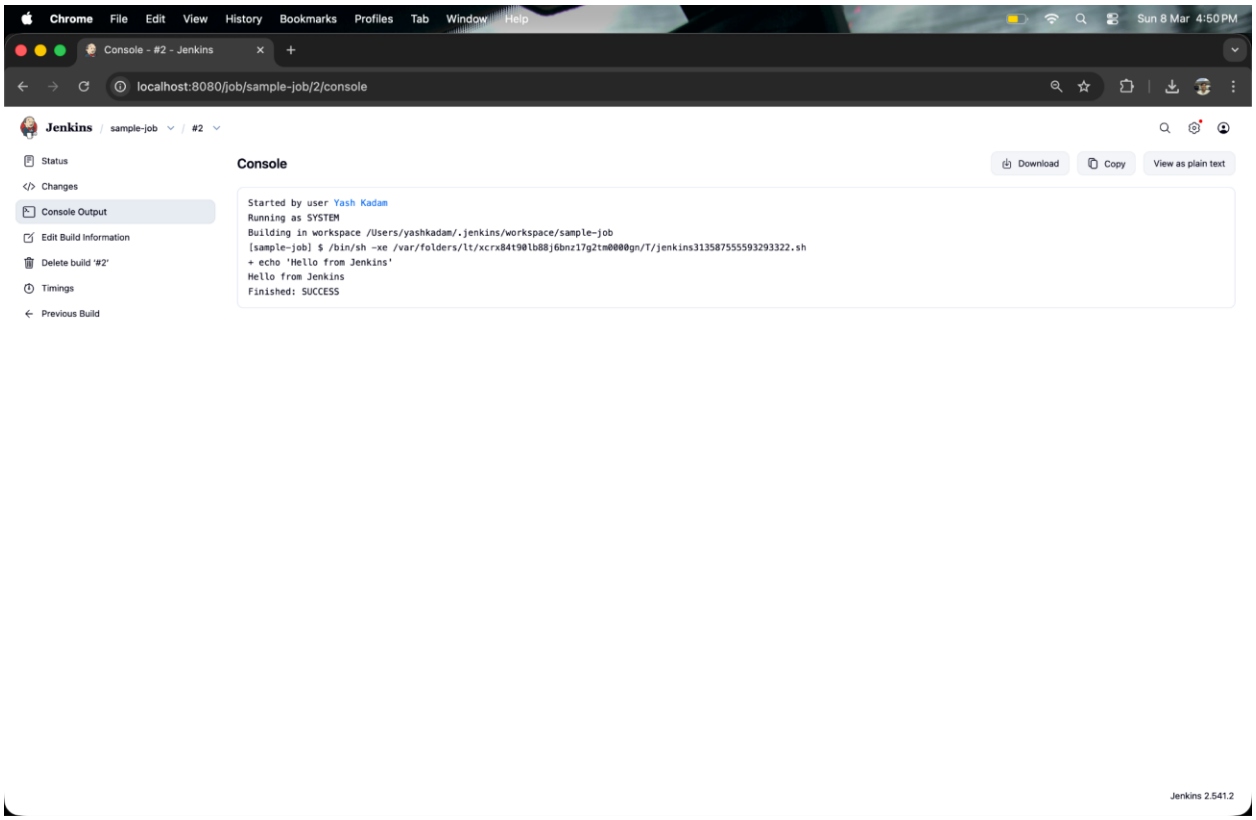

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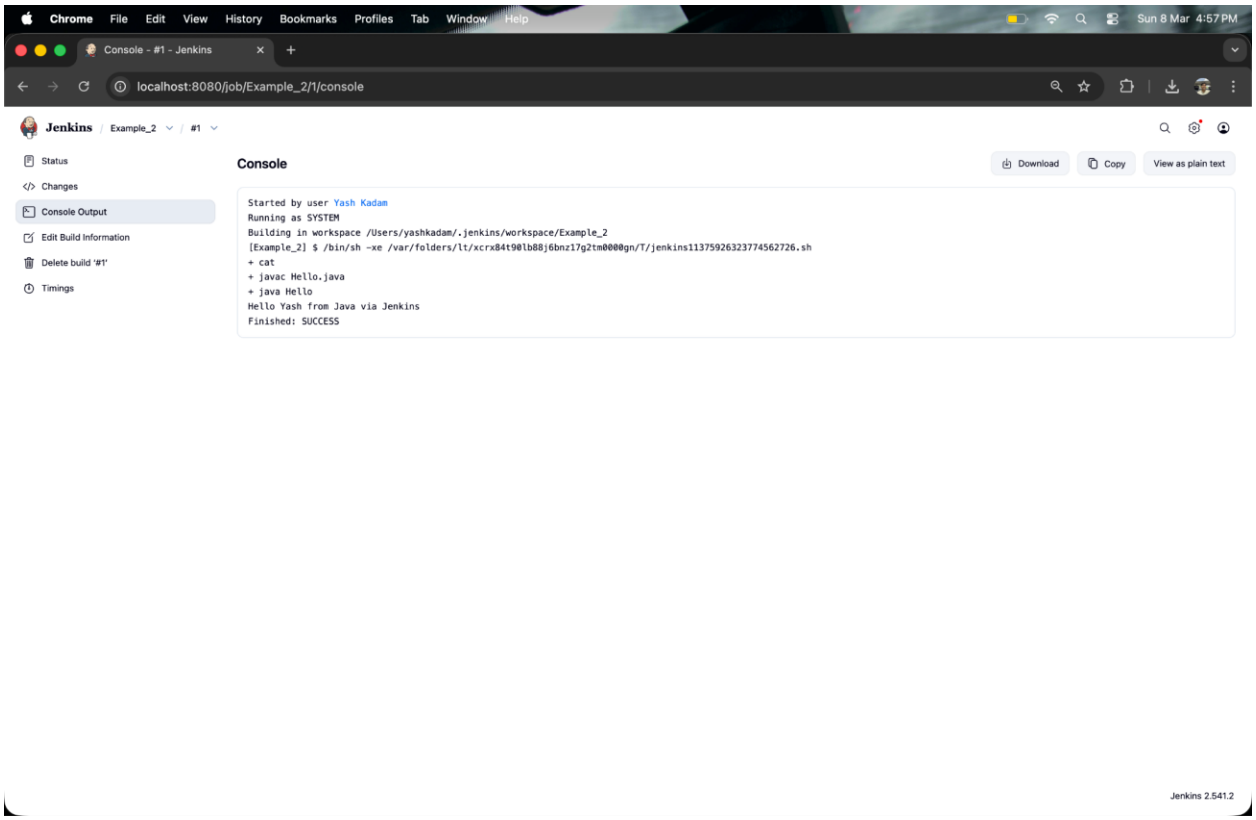

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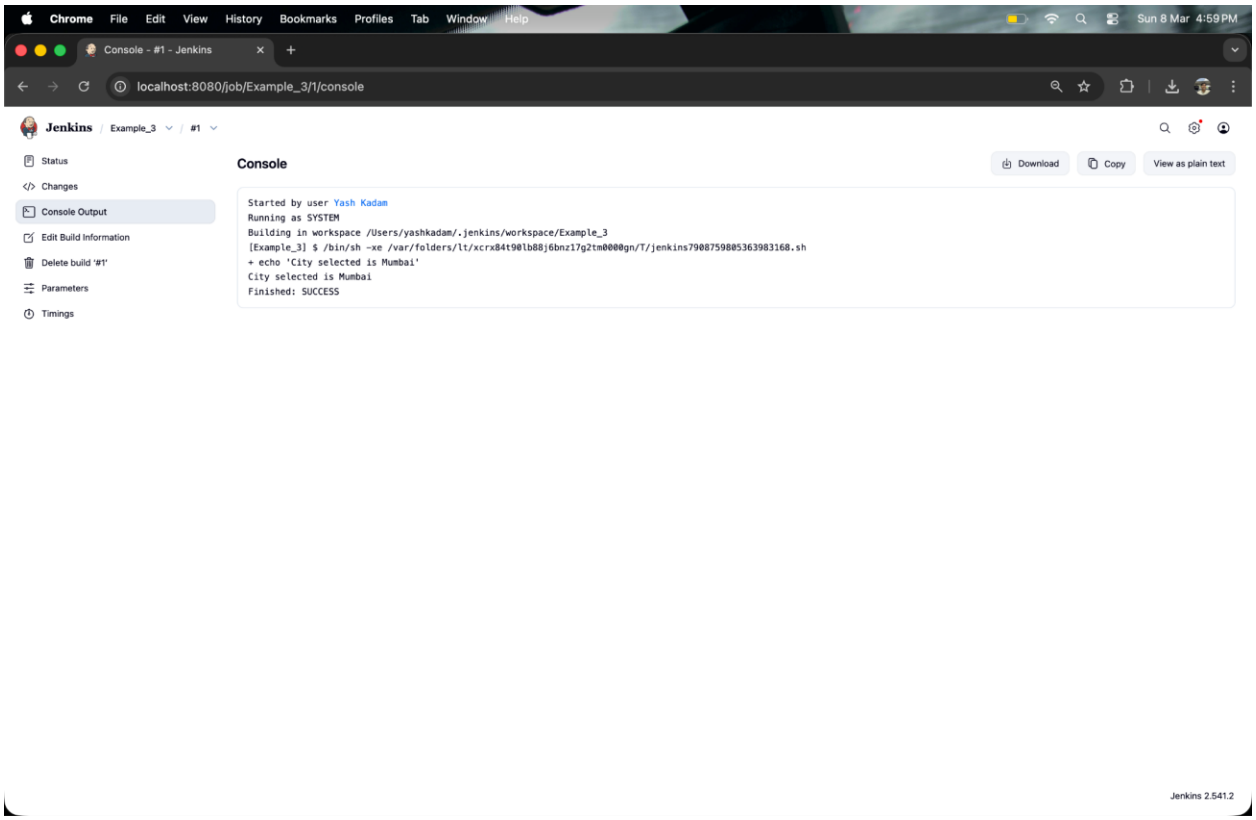

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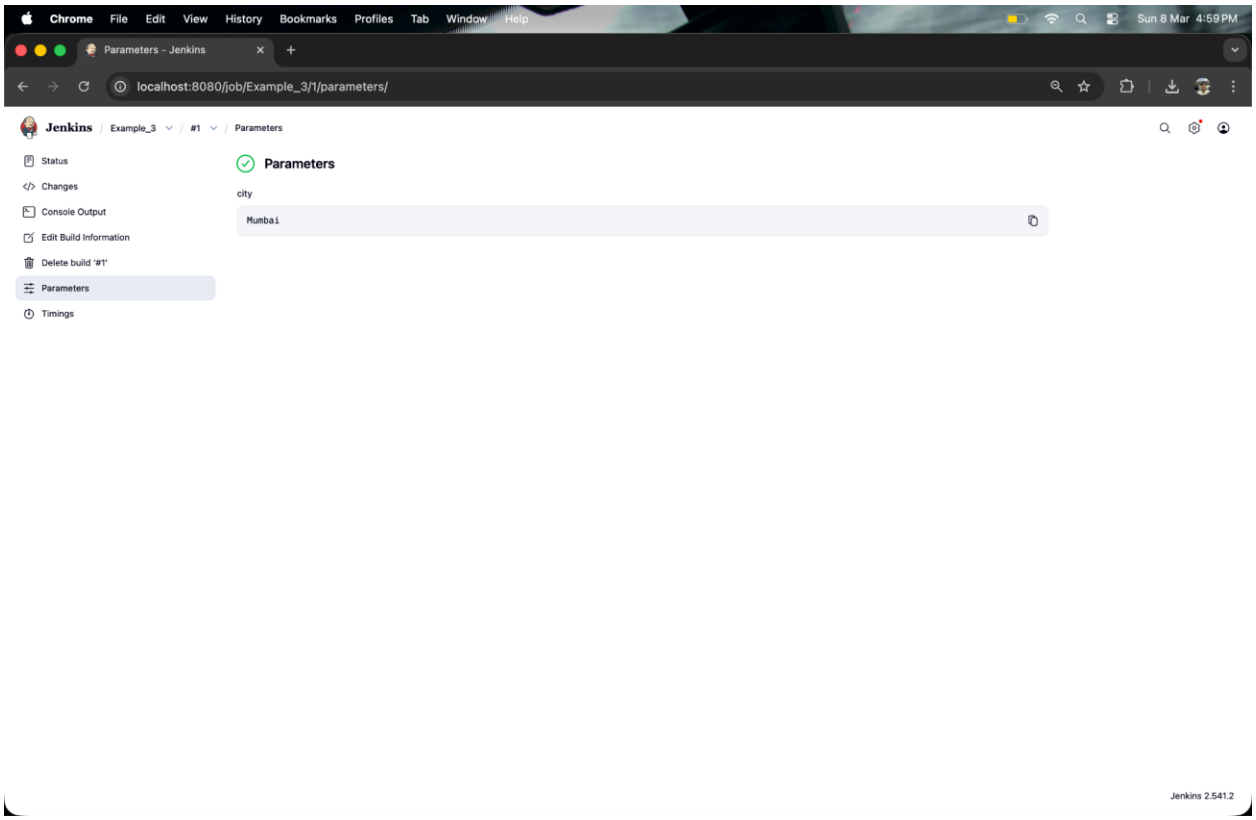

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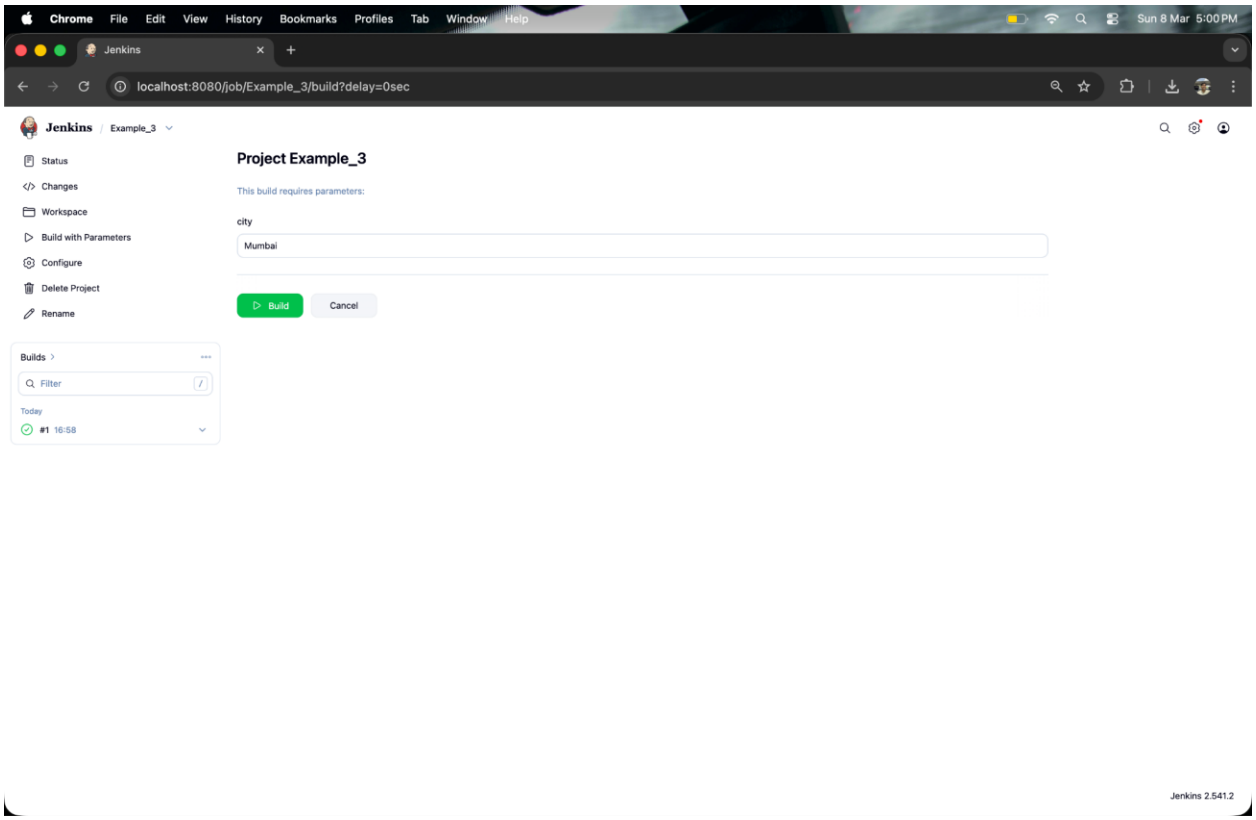

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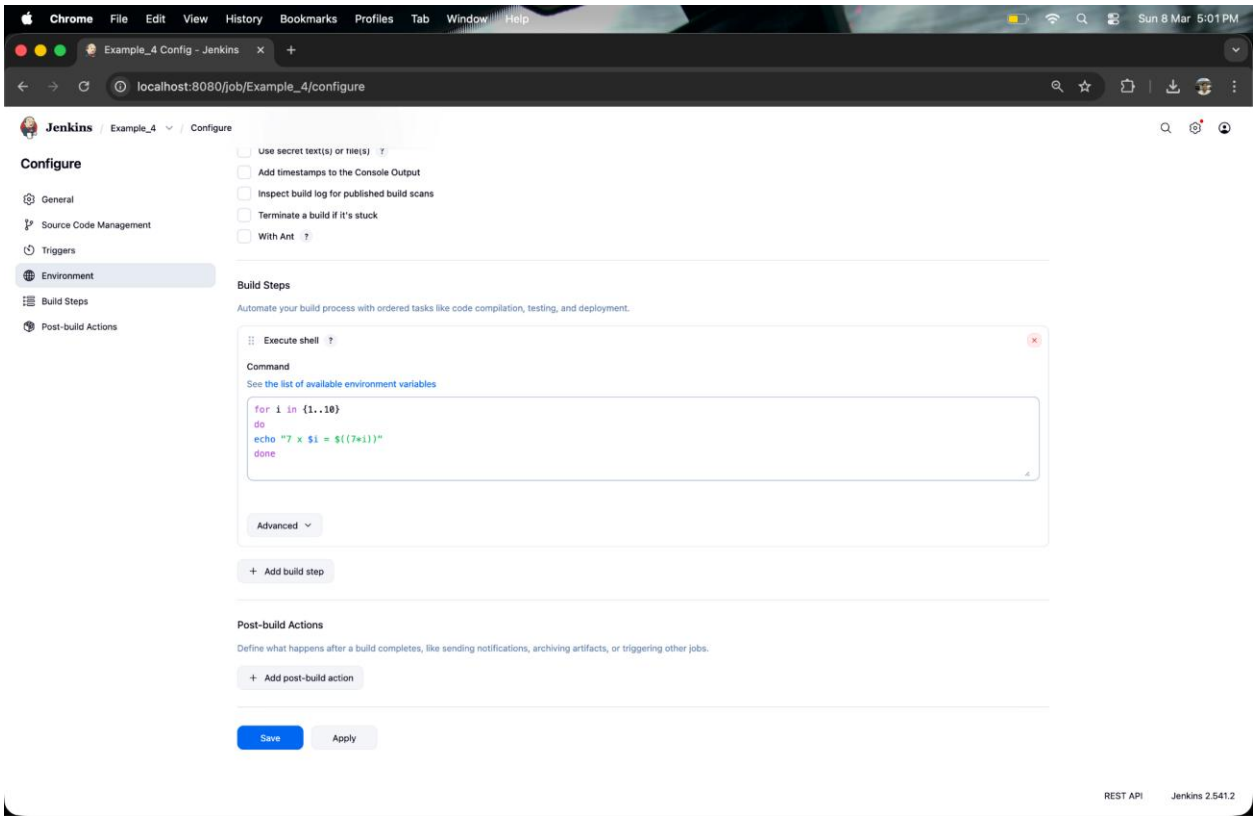

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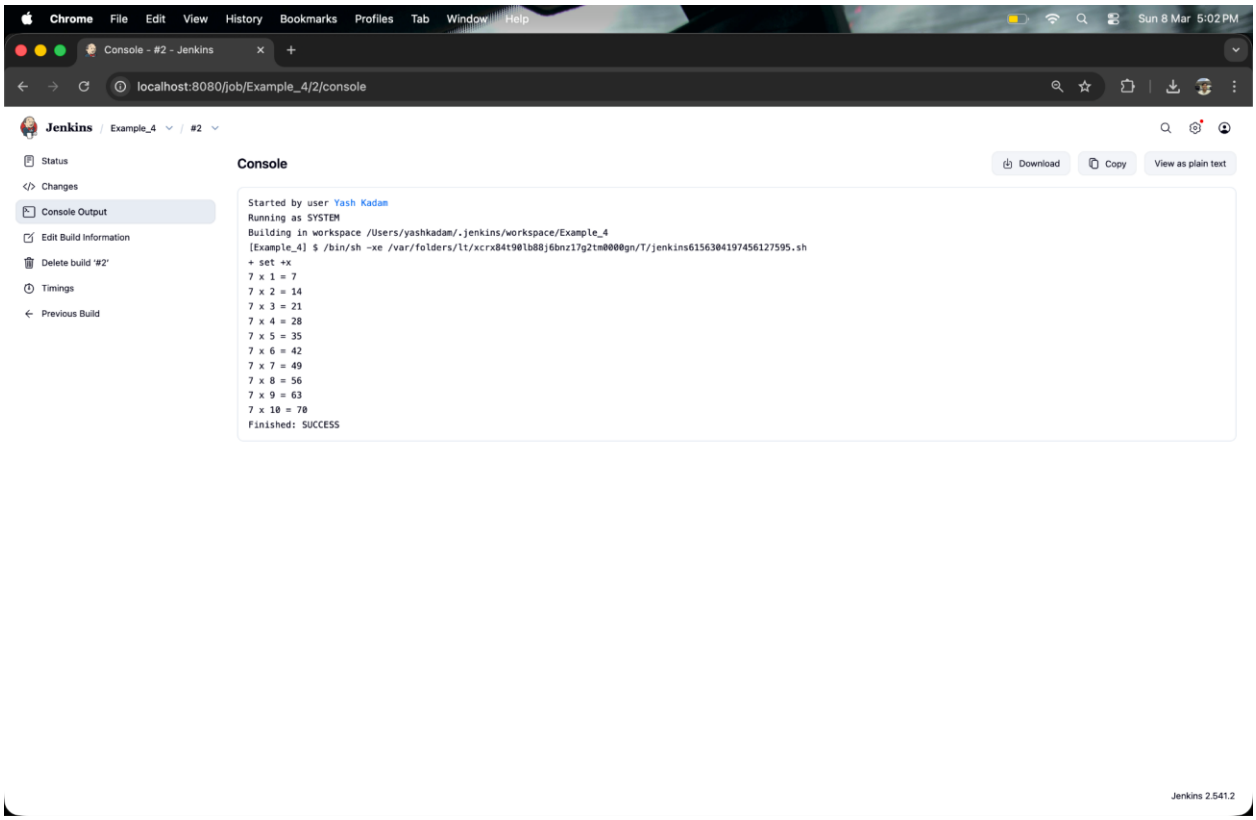

Conclusion:

Successfully Implemented Continuous Integration, install and configure Jenkins with
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